

4.3 How do design engineers use different approaches to design thinking to support the development of design ideas?

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| a. | Awareness of different strategies, techniques and approaches to explore, create and evaluate design ideas, including: <ul style="list-style-type: none">• iterative designing• user-centred design• circular economy• systems thinking. | |
| b. | The importance of collaboration to gain specialist knowledge from across subject areas when delivering solutions in the design and manufacturing industries. | |
| c. | Understand how design engineers use system design processes to define and develop systems that satisfy specified requirements of users using the three sub-tasks of: <ul style="list-style-type: none">i. user-interface designii. data designiii. process design. | |
| d. | Understand how design teams use different approaches to project management when faced with large projects, such as critical path analysis, scrum and six sigma. | |

5. Material and component considerations

Considerations

*Maths &
Science*